

Investigation of environmentally friendly nutrient replenishment options for grasslands with saline soil conditions in Karcag

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SUMMARY

One of the key issues in sustainable grassland management is ensuring adequate nutrient replenishment in an environmentally friendly manner. The aim of this study was to investigate the effects of different doses of organic manure and wood ash on the productivity and botanical composition of a saline grassland. During the experiments, we measured soil pH, electrical conductivity, moisture content, CO₂ emissions, and the fresh biomass yield and species composition of the plant community. Based on the experiments, manure had a positive effect primarily on increasing fresh biomass yield, while wood ash had a positive effect on reducing soil acidity. Both treatments had a positive effect on microbial activity and soil structure. Changes in species composition suggest that the interventions may have a long-term beneficial effect on grassland biodiversity.

Keywords: wood ash, manure, saline grassland, soil factor, botany, fresh biomass yield

ÖSSZEFOGLALÁS

A fenntartható gyepgazdálkodás egyik legfontosabb kérdése a megfelelő tápanyag-utánpótlás környezetbarát módon történő biztosítása. E tanulmány célja az volt, hogy megvizsgálja a különböző adagú szerves trágya és fahamu hatását egy szikes gyep termelékenységére és növénytani összetételére. A kísérletek során megmértük a talaj pH-értékét, elektromos vezetőképességét, nedvességtartalmát, a CO₂-kibocsátást, valamint a növénytársulás zöld biomasszáját és fajösszetételét. A kísérletek alapján a trágya elsősorban a biomassza-hozam növelésére gyakorolt pozitív hatást, míg a fahamu a talaj savasságának csökkentésére volt kedvező hatással. Mindkét kezelés pozitív hatással volt a mikrobiális aktivitásra és a talajszerkezetre. A fajösszetételben bekövetkezett változások arra utalnak, hogy a beavatkozások hosszú távon kedvező hatással lehetnek a gyep biodiverzitására.

Kulcsszavak: fahamu, trágya, szikes gyep, talajtényező, botanika, zöld biomassza-hozam

INTRODUCTION

Organic soil conditioners such as manure and wood ash have long been used to improve soil fertility and structure. Wood ash, a by-product of biomass combustion, provides essential nutrients and can raise soil pH due to its alkaline nature, while manure improves soil structure and nutrient supply capacity

primarily through its organic matter content (Brady and Weil, 2008; Gianello and Bremner, 2008). Such organic amendments not only improve the physical properties of the soil, but also increase the biological availability of nutrients and stimulate microbial activity (Lal, 2015; Brady and Weil, 2008). Wood ash, in particular, adds significant amounts of potassium, calcium, and micronutrients to the soil, while manure increases soil organic matter (SOM) content, enhances water retention capacity, and supports the functioning of beneficial microorganisms in the soil.

The effects of wood ash on soil have been investigated in numerous studies. Costa et al. (2020) point out that although wood ash is alkaline, its high inorganic nutrient content makes it a more environmentally friendly alternative to lime hydrate for liming and potassium nitrate for potassium supplementation. However, the choice of dosage is a critical factor. According to a study by Reed et al. (2017), hardwood ash produced at 870 °C had no negative effect on the structure of the soil microbial community at doses of up to 571 kg/ha, and thus did not affect organic matter mobilization. In an experiment conducted by Qin et al. (2017) on clayey sandy soil in Denmark, the application of wood ash at a dose of up to 17.4 t/ha did not cause damage to the soil mesofauna, but significantly increased the soil pH. Ondrasek et al. (2021) observed a two-unit increase in pH when wood ash was applied to acidic, hydromorphic soils and showed that the amendment increased the soil content of P, K, Ca, Mg, Zn, Mn, and Cu, while reducing cadmium mobility. Blonska et al. (2023) applied wood ash with a pH of 12 at doses of three (3.0, 4.5 and 6.0 t/ha) of wood ash with a pH of 12 on acidic brown forest soils in southern Poland, which increased the soil's chemical properties by 1.5 pH units regardless of the dose and also had a significant effect on the composition of fungal communities.

With regard to grassland fertilization, de Sainte Marie's (2014) studies on high conservation value grasslands in France showed that moderate organic fertilization increases species richness, exceeding the values of untreated areas. Environmentally friendly fertiliser management can increase the conservation and agricultural value of grasslands in terms of both flora (Verrier and Kirkpatrick, 2005; Chytry et al., 2015) and fauna (D'Aniello et al., 2011). Huyghe et

al. (2014) emphasize that there is a growing demand for ruminant meat in the European Union, which also increases demand for forage, so chemical-free, environmentally friendly yield increases in grasslands are crucial. Although several authors in Hungarian literature (Takáts, 1954; Milkovich, 1962; Petrányi, 1963; Balla, 1964) previously considered grassland fertilization to be impractical, more recent experimental results show positive effects. Szopkó and Barcsák (1992) applied 20 t/ha of organic manure during the planting of *Festuca arundinacea* (Schreb.) and achieved a 26% higher dry matter yield (6.22 vs. 4.94 t/ha) compared to the control. Based on Bánszki (1993), 15-35 t/ha of autumn manure application is recommended, which can be expected to increase yields by 30-50%. In Kádár et al. (2007)'s experiment in Bakonszeg, 10 t/3 years of sheep manure resulted in 1.5 t/ha of additional green grass and increased biodiversity. Csizi and Monori (2007, 2008) achieved a high proportion of leguminous cover by applying 20 t/ha of composted sheep manure. Kovács et al. (2010) recommended using organic manure-based compost in solonchak soils to enhance soil life (CO₂ emissions) and moisture retention. According to the experiences of Török et al. (2013), fertilization increased the proportion of useful grasses and legumes, while suppressing the aggressively spreading reed canary grass. Díaz and Csizi (2017) and Díaz et al. (2018) confirmed the yield-increasing and species richness-promoting effects of composted organic sheep manure on near-natural grasslands with Karcag meadow solonetz soil conditions.

The aim of the study is, on the one hand, to examine the effects of wood ash and organic manure on soil pH, electrical conductivity, microbiological activity, and fresh biomass development on a grassland in the Great Hungarian Plain. On the other hand, our objective is to compare the effectiveness of applying the two amendments separately, with particular regard to optimizing productivity and reducing environmental impact. As a third objective, the research contributes to developing sustainable grassland management strategies that strike a balance between ecological and economic considerations.

During the study, we formulated the following hypotheses:

- Organic amendments increase soil moisture content and improve soil structure.
- Wood ash raises soil pH and reduces its salt content.
- The amendments stimulate microbial activity and increase soil CO₂ emissions.
- Soil amendments enhance biomass and protein production.
- The different treatments influence the species composition of the grassland.

MATERIALS AND METHODS

Our experiment was conducted in 2023 at the Hungarian University of Agricultural and Life Sciences, Agrárcsoport Kft., on a 17-hectare grassland

area with meadow solonetz soil (47°17'20.4"N 20°55'14.7"E).

The experimental plots were established within this area, as illustrated in the map below (Figure 1). The area forms a transition between the *Achilleo-Festucetum pseudovinae* and *Alopecuretum pratensis* grassland communities and is subject to Natura 2000 environmental protection regulations (Government Decree 269/2007 (X.18.); July 10, 2025).

Figure 1: Layout of the plots in the soil reclamation experiment

A20-1	A10-1	A30-1	A0-1	A10-2	A30-2	A0-2	A20-2	A30-3	A0-3	A20-3	A10-3
M15-1	M10-1	M20-1	M0-1	M10-2	M20-2	M0-2	M15-2	M20-3	M0-3	M15-3	M10-3

Legends: A0 and M0 = untreated control; A10, A20, A30 = wood ash application of 10, 20, 30 g m⁻²; M10, M15, M20 = manure application of 10, 15, 20 t ha⁻¹; last numbers (1-3) indicate the replications(1)

1. ábra: A talajrekultivációs kísérlet parcelláinak elrendezése

Jelmagyarázat: A0 és M0 = kezetlen kontroll; A10, A20, A30 = fahamu kijuttatása 10, 20, 30 g m⁻² mennyiségben; M10, M15, M20 = szerves trágya kijuttatása 10, 15, 20 t ha⁻¹ mennyiségben; az utolsó számok (1-3) az ismétléseket jelzik(1)

The first experiment was set up in the fall of 2022 with four ash treatment levels: 0 (control), 10, 20 and 30 dk/plot (equivalent to 0, 3.3, 6.7, and 10 t/ha, respectively). Each treatment was arranged in three replicates with a net plot size of 30 m². The second experiment, also initiated in the fall of 2022, tested organic manure at rates of 0 (control), 10, 15 and 20 t/ha. This experiment was also organized in three replicates, using a net plot size of 30 m² for each parcel. In both experiments, 0.5 m access paths were maintained between the plots, and each parcel received only one specific treatment type (H0–H3 for ash, T0–T3 for manure). Prior to setting up the experiments, the average soil sample was tested by the accredited laboratory of the Central Laboratory Center of the Hungarian University of Agricultural and Life Sciences, with the following results: pH (KCl) 4.51, soil plasticity of Arany (K_A) 49, humus 4.83 m/m%, nitrogen content 17.1 mg/kg, phosphorus pentoxide content 46.8 mg/kg, potassium oxide content 482 mg/kg, sodium content 484 mg/kg (report number: t 728-731-2023).

The average precipitation over 50 years is 503 mm. The meteorological data for the area relevant to the experiment are presented in Table 1 and were collected by the Hungaromet Karcagi automatic station (WMO code: 55405).

The soil samples were taken from the soil of the wood ash and organic manure application plots on two occasions (June 13 and October 19, 2023), in parallel with carbon dioxide concentration measurements. From each plot, 3 soil samples were collected from the top 0-10 cm layer and placed in metal sample containers, which were covered with plastic lids to preserve the actual moisture content of the soil.

Table 1

**The meteorological data of the experiment
(Karcag, January 2022 – October 2023)**

Date(1)	Monthly rainfall (mm)(2)	Monthly average temperature (°C)(3)
January 2022	60.1	4.4
February 2022	6.8	2.6
March 2022	34.6	7.4
April 2022	39.6	9.6
May 2022	49.9	16.6
June 2022	51.6	20.3
July 2022	37.7	23.7
August 2022	50.9	23.7
September 2022	42.6	20.7
October 2022	31.7	14.6
November 2022	102.5	6.4
December 2022	64.3	2.9
January 2023	23.1	1.2
February 2023	6.0	8.4
March 2023	13.2	9.7
April 2023	30.0	14.2
May 2023	34.2	18.0
June 2023	48.5	22.5
July 2023	33.8	25.6
August 2023	7.9	25.7
September 2023	70.6	18.8
October 2023	44.2	11.5

1. táblázat: A kísérlet meteorológiai adatai (Karcag, 2022. január – 2023. október)

Dátum(1), havi csapadék (mm)(2), havi középhőmérséklet(3)

The soil samples were then dried in a drying oven at 95 °C to determine their actual moisture content (m/m%) by weighing the sampling containers and calculating it from the change in soil mass. The pH and electrical conductivity (EC) of the soil were measured using an InoLab pH meter and an InoLab EC meter, respectively, with 3 instrumental measurement repetitions per sample. Soil CO₂ emissions were examined using a framework method developed at the

Karcag Research Institute of the University of Debrecen (Kovács, 2014), where a Testo 535 infrared gas analyzer was used to measure CO₂ concentration. The phytosociological survey of the plant community was carried out using the Balázs quadrat method on April 8, 2023, and May 9, 2024 (Balázs, 1960). Plant species were classified according to Király (2009). After the botanical survey, the phytomass was cut, and the green yield of the grassland (fresh biomass yield) was measured.

The data collected in the experiments were recorded and summarized, and the results were processed and evaluated using Microsoft® Office Excel. We used one-way analysis of variance (ANOVA) to analyze the data. For statistical evaluation, we used the p-value from the elements of variance analysis at a 5% significance level.

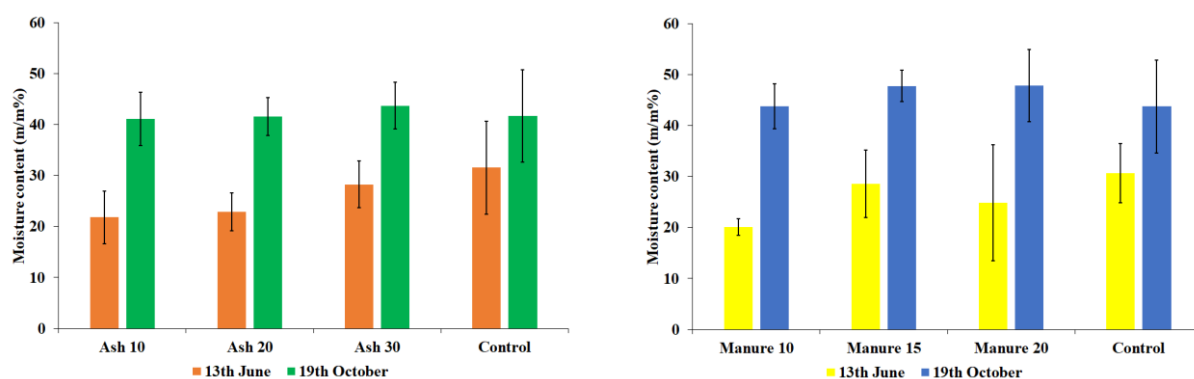
RESULTS

Soil moisture changes under different nutrient treatments

During the study period (June 13 to October 19), soil moisture content increased intensely under all treatments, which may be indicative of typical autumn moisture conditions (Figure 2). In the case of ash treatments, the moisture content ranged from 21.8 to 31.5 w/w% in June, while in October it increased to 41.1-43.7 w/w%. The areas treated with ash did not show significant results either in June ($F=2.74$; $p=0.11$) or October ($F=0.16$; $p=0.92$).

In the manure treatments, the June values ranged from 20.1 to 30.6%, and the October values ranged from 43.8 to 47.9 w/w%. The 15 t/ha and 20 t/ha manure treatments showed the highest moisture content in October, reaching 47.8-47.9 w/w%. This may indicate that manure treatment is better at retaining moisture in the soil, especially in the autumn, which may be an important factor in plant water supply and soil microbiological activity. The areas treated with manure did not show significant results in June ($F=1.23$; $p=0.36$) or October ($F=0.41$; $p=0.75$).

Figure 2: Actual average soil moisture contents in the ash and manure application experiment (n=3)



2. ábra: A hamu- és trágyakijuttatási kísérletben mért tényleges átlagos talajnedvességtartalom (n=3)

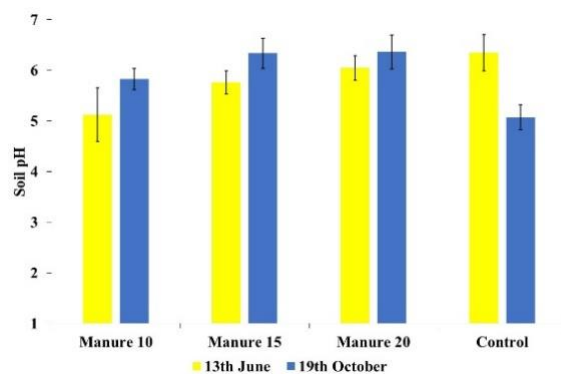
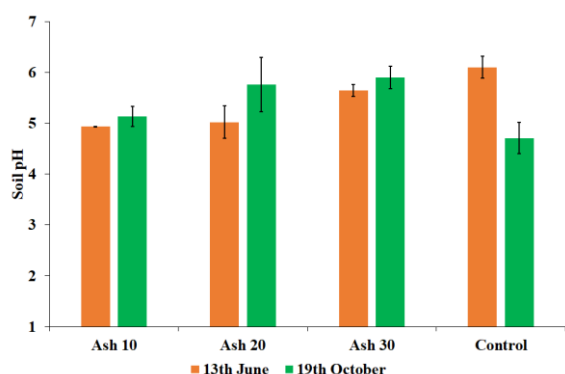
Changes in soil pH as a result of different nutrient treatments

Soil pH is an important factor in terms of nutrient availability and microbiological activity (Figure 3). During the study period (June 13 to October 19), the following changes were observed: the pH of ash treatments generally increased. The 10 dk/plot ash treatment increased from 4.9 to 5.1, the 20 dk/plot ash treatment also increased from 5.0 to 5.8, while the 30 dk/plot ash treatment showed only a slight increase from 5.6 to 5.9. However, in the ash control plot, a decrease in pH was observed from 6.1 to 4.7. We also observed an increase in pH in the manure

treatments. The areas treated with ash showed significant results in June ($F=22.36$; $p=0.0003$) and October ($F=7.90$; $p=0.01$).

In the 10 t/ha manure treatment, it increased from 5.1 to 5.8, in the 15 t/ha manure treatment from 5.8 to 6.3, and in the 20 t/ha manure treatment from 6.0 to 6.4. In the manure control plot, it decreased from 6.4 to 5.1. The manure and ash treatments generally had a slight alkalizing effect on the soil, while the pH of the control treatments decreased, indicating acidification. The areas treated with manure showed significant results in June ($F=6.28$; $p=0.02$) and October ($F=14.18$; $p=0.001$).

Figure 3: Actual average soil pH in the ash and manure application experiment (n=3)



3. ábra: A talaj tényleges átlagos pH-értéke a hamu- és trágyakijuttatási kísérletben (n=3)

Soil electrical conductivity (EC) under different treatments

Electrical conductivity (EC) indicates the amount of dissolved ions in the soil, which is directly related to the soil's nutrient content and salt concentration (Figure 4). For the treatments studied, the EC values on June 13 and October 19 were as follows: for the ash treatments, EC ranged from 323.7 to 392.7 $\mu\text{S}/\text{cm}$ in June, and from 328.7 to 392.3 $\mu\text{S}/\text{cm}$ in October. The EC increased in the ash control plot (from 344.7 $\mu\text{S}/\text{cm}$ to 370.7 $\mu\text{S}/\text{cm}$). The ash-treated areas did not show significant results either in June ($F=1.71$; $p=0.24$) or in October ($F=0.68$; $p=0.59$).

Higher EC values were measured for the manure treatments, ranging from 348.0 to 406.7 $\mu\text{S}/\text{cm}$ in June and changing to 299.0 to 434.0 $\mu\text{S}/\text{cm}$ in October. In the 20 t/ha and 10 t/ha manure treatments, a significant increase in EC was observed in October, indicating nutrient accumulation in the soil. The manure-treated areas did not show significant results in June ($F=0.20$; $p=0.89$) or October ($F=1.05$; $p=0.42$).

Fertilization increased the soil nutrient level (EC), whereas changes in the ash treatments were smaller; in fact, a decrease was observed in October for the 30 dk/plot ash treatment.

Changes in soil CO₂ emissions under different nutrient treatments

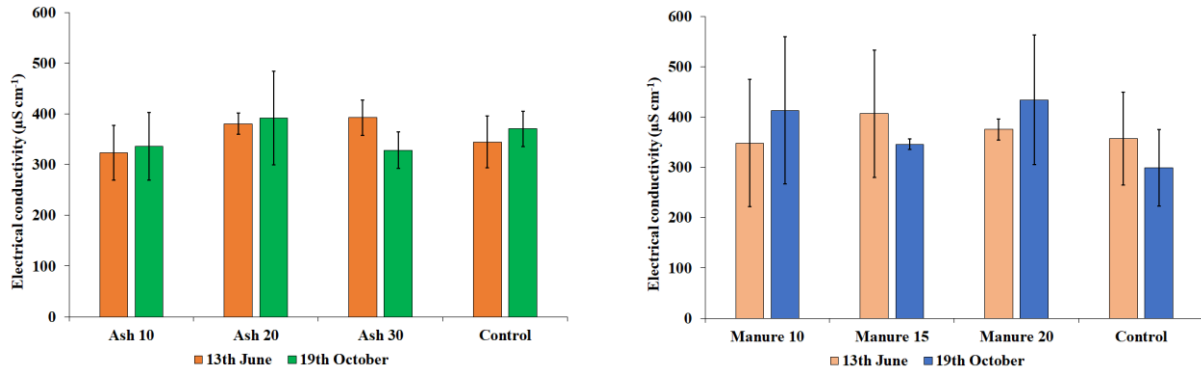
Soil respiration (CO₂ emissions) can be an important indicator of microbial activity and ecosystem health (Figure 5). The CO₂ emission values ($\text{g}\times\text{m}^{-2}\times\text{h}^{-1}$) measured during the study were as follows in June and October: in the ash treatments, CO₂ emissions ranged from 0.169 to 0.243 $\text{g}\times\text{m}^{-2}\times\text{h}^{-1}$ in June, then decreased to 0.130-0.161 $\text{g}\times\text{m}^{-2}\times\text{h}^{-1}$ in October. The highest emissions were observed in the 30 dk/plot ash treatment in June (0.243 $\text{g}\times\text{m}^{-2}\times\text{h}^{-1}$), which decreased to 0.147 $\text{g}\times\text{m}^{-2}\times\text{h}^{-1}$ by October. The CO₂ emissions of the ash control plot were 0.213 $\text{g}\times\text{m}^{-2}\times\text{h}^{-1}$ in June, but decreased to 0.130 $\text{g}\times\text{m}^{-2}\times\text{h}^{-1}$ by October. The ash-treated areas did not show significant results in June ($F=0.51$; $p=0.67$) or October ($F=0.10$; $p=0.96$).

In the case of manure treatments, CO₂ emissions were higher than in the case of ash, indicating more intense soil life and nutrient cycling. By October, CO₂ emissions had decreased significantly, probably due to cooler weather. In June, they ranged from 0.226 to 0.402 $\text{g}\times\text{m}^{-2}\times\text{h}^{-1}$, and by October they had decreased to 0.111-0.186 $\text{g}\times\text{m}^{-2}\times\text{h}^{-1}$.

The highest value was achieved by the 10 t/ha manure treatment in June ($0.402 \text{ g} \times \text{m}^{-2} \times \text{h}^{-1}$), while in October the 15 t/ha and 20 t/ha manure treatments had the highest values (0.186 and $0.169 \text{ g} \times \text{m}^{-2} \times \text{h}^{-1}$). The values for the manure control plot were lower:

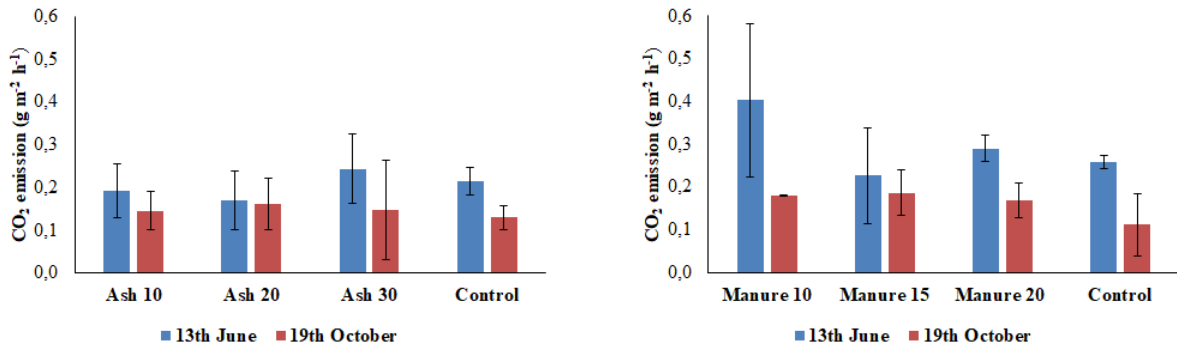
$0.259 \text{ g} \times \text{m}^{-2} \times \text{h}^{-1}$ in June and $0.111 \text{ g} \times \text{m}^{-2} \times \text{h}^{-1}$ in October. CO_2 emissions were highest during the summer period, which can be explained by increased microbial activity.

Figure 4: Actual average soil EC in the ash and manure application experiment (n=3)



4. ábra: A hamu- és trágyakijuttatási kísérletben mért tényleges átlagos talaj-elektromos vezetőképesség (n=3)

Figure 5: Actual average of CO_2 emission values in the ash and manure application experiment (n=3)



5. ábra: A hamu- és trágyakijuttatási kísérletben mért CO_2 -kibocsátási értékek tényleges átlaga (n=3)

Fresh biomass yields under different nutrient treatments

Changes in fresh biomass yield quantity were measured in three replicates for different manure and ash treatments (Figure 6). Increasing the manure dose generally increased biomass yield, with the 20 t/ha manure treatment achieving the highest average value (1.02 t/ha). In the case of ash treatments, the 10 and 20 dkg/plot treatments yielded the best biomass values (0.95 t/ha , 0.97 t/ha), while the 30 dkg/plot treatment (0.77 t/ha) resulted in lower biomass compared to the control (0.81 t/ha). Manure treatments had a stronger positive effect on fresh biomass growth, while among ash treatments, medium doses yielded better results, with the highest ash dose showing a decrease. The variance analysis showed no significant correlation in either the ash-treated ($F=1.06$; $p=0.40$) or manure-treated ($F=1.64$; $p=0.23$) areas.

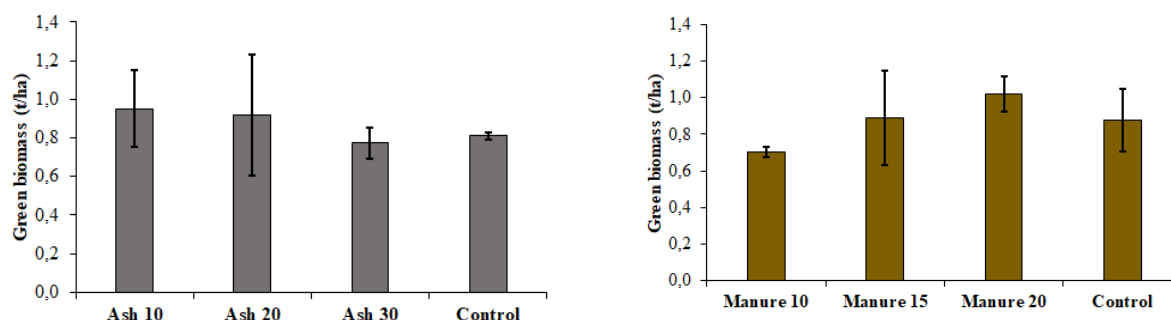
Changes in plant species coverage as a result of different treatments

Compared to the baseline survey in 2023, the nutrient replenishment treatments applied in 2024 had a significant impact on the coverage of several plant species (Figure 7). The species composition and coverage observed on the treated plots differed significantly from the control area, indicating the species-specific effect of the treatments. *Alopecurus pratensis* (L.) increased from 25% coverage in 2023 to 55.32% (ash treatment) and 52.93% (manure treatment) in 2024, while in the control it was only 21.51%. This suggests that this species responds particularly well to nutrient replenishment, with more than double the growth observed in the treated plots. *Plantago lanceolata* (L.) also responded favorably to both treatments, as its coverage increased significantly compared to the baseline (0.78%): 11.5% after ash treatment and 9.38% after manure treatment, while the control group showed moderate growth (14.84%).

In the case of several species, their coverage only appeared or increased significantly as a result of the treatments. *Poa pratensis* (L.), *Potentilla argentea* (L.), *Vicia tetrasperma* (L.) Schreb.) and *Ranunculus acris* (L.) were present only in the areas treated with manure or ash, while they were absent in the control and in the baseline survey. These species may be potential indicators of positive nutrient effects. In contrast, the coverage of *Podospermum canum* (C. A. Mey.) was lower in the treated areas, while it was higher in the control. *Festuca rupicola* (Heuff.) showed more complex responses: although it was also present in the treated areas, its coverage did not always increase compared to the control, which may indicate

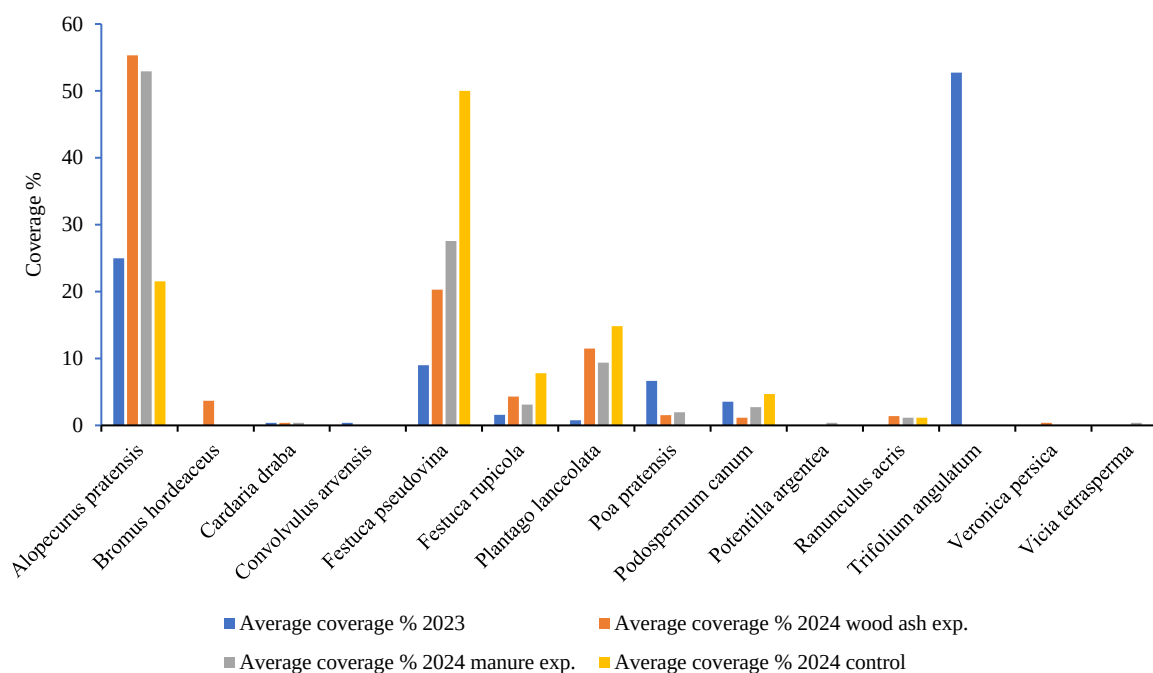
competition- and habitat-dependent behavior. *Festuca pseudovina* (Hack. ex Wiesb.) was dominant in the control area (50%), while it showed significantly lower coverage in the fertilized and ash-treated plots (27.54% and 21.31%, respectively). This may indicate that these species are more competitive in nutrient-poor environments. In the case of *Trifolium angulatum* (Waldst. et. Kit.), the change was negative, as the species dominated the area with 52.73% coverage during the baseline survey, but disappeared completely after nutrient replenishment. This may indicate that this species is sensitive to elevated nutrient levels and survives only in low-productivity environments.

Figure 6: Total fresh biomass yields in the manure and ash application experiment in 2023



6. ábra: A teljes zöld biomassza-hozam a 2023-as trágyázási és hamuzásos kísérletben

Figure 7: Average coverage percentage of plant species in 2023 and under different soil treatments in 2024 (wood ash, manure, and control)



7. ábra: A növényfajok átlagos borítottsága 2023-ban, valamint különböző talajkezelések esetén 2024-ben (fahamu, trágyázás és kontroll)

DISCUSSION

The results of the experiment examining the soil-improving effects of organic fertilization and wood ash confirm, in several respects, the experiences previously described in domestic and international literature. The most significant effect of wood ash application was an increase in soil pH, especially at medium doses. This is consistent with the observations of Brady and Weil (2008) and Blonska et al. (2023), who found that wood ash can significantly improve the chemical properties of acidic soils through its alkalizing effect. Similar results were reported by Ondrasek et al. (2021), who observed a 2-pH-unit increase in hydromorphic soils treated with wood ash. However, it is important to note that yield decreased at the highest dose (20 t/ha), confirming the findings of Costa et al. (2020) and Reed et al. (2017) regarding the potential negative effects of excessive wood ash doses.

The increase in biomass yield content was mainly due to organic fertilization. The 20 t/ha dose proved particularly effective, consistent with the results of Csízi and Monori (2007, 2008), Török et al. (2013), and Bánszki (1993). Fertilization had a positive effect on soil water retention capacity, soil structure, and microbial activity, which was reflected in higher CO₂ emissions and more vigorous soil life during the growing season (Lal, 2015; Huyghe et al., 2014). Similar findings were reported by Csízi et al. (2023a, b), who showed that fertilization treatments improved moisture content, promoted soil life, and contributed to the maintenance of species richness in the long term.

Wood ash treatment not only increased pH, but also had a positive effect on biomass yield and plant community composition. According to an experiment by Asbolt et al. (2024) found that wood ash increased the proportion of *Poa* and *Alopecurus* species and improved phytomass quantity, which supports the trend observed in the present study. These effects are particularly important in the reconstruction of degraded, low-yield grasslands and in near-natural management.

Changes in species composition showed an increase in the coverage of useful grassland plants, especially grasses and legumes, while competitive weed species were suppressed.

This trend is consistent with the findings of Kovács et al. (2010), Kádár et al. (2007) and Díaz et al. (2018). This type of structural shift also increases the species richness and ecological stability of grasslands, so the right combination of fertilization and ash application can contribute to the multifunctional use of grasslands in the long term, including nature conservation, forage production, and climate resilience.

Wood ash and organic manure, both separately and in combination, positively affect grassland productivity, soil chemical and biological parameters, and increase the conservation value of grasslands. Optimizing dosage, application timing, and consideration of soil type and vegetation status are key to achieving the desired ecological and economic effects. Our results confirm that both interventions can be valuable tools in the implementation of environmentally friendly, chemical-free grassland management strategies.

CONCLUSION

Our study clearly confirmed that the application of wood ash and organic manure had a positive effect on the chemical and biological properties of the grassland soil and on the yield of fresh biomass. Wood ash proved to be particularly effective in improving the pH of acidic soils and replenishing microelements. Organic manure increased grassland productivity by improving soil structure and moisture retention, which likely facilitated more favorable conditions for soil biological processes.

Fertilization also positively affected the grassland species composition, increasing the cover of beneficial grassland species. We also obtained positive results in suppressing competing species (*Trifolium angulatum*, *Podospermum canum*). The most favorable results were obtained with the application of 20 t/ha of organic manure or a medium dose (20 dkg/plot) of wood ash.

Based on the results, it can be concluded that organic manure and wood ash treatments can be effective tools in a sustainable grassland management strategy. The use of these natural nutrient-replenishing materials can contribute to the sustainable use of grasslands, increase their productivity, and preserve their conservation value in the long term.

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